

hand-calculated values given above. However, from the criteria given in Eq. (24.9), the op-amp will clearly be unstable. *We want the open loop gain of the op-amp to be much less than one when the phase shift is 180° .* Looking at Fig. 24.10, we effectively want to shift the lower frequency pole (f_1) downwards in frequency. At the same time, we want to move the higher frequency pole, f_2 , higher in frequency (we want to *split the poles*). Remember this (pole-splitting) was the point of adding the compensation capacitor to an amplifier back in Ch. 21. Let's use Eq. (21.72) to select C_c . Looking at Fig. 24.10, we see that the phase shift is -100° at 10 MHz, so let's set the unity gain frequency to this value (in an attempt to make sure that the open loop gain is well under one when the phase shift is 180°).

$$f_{un} = \frac{g_{m1}}{2\pi C_c} = \frac{150 \mu A/V}{2\pi \cdot C_c} = 10 \text{ MHz} \rightarrow C_c = 2.4 \text{ pF} \quad (24.10)$$

The simulation results are seen Fig. 24.11. The unity-gain frequency, f_{un} , is close to 10 MHz. However, while the lower frequency moved downwards and the higher frequency pole moved upwards, the zero moved down to unity-gain frequency, see Eq. (21.63). This causes the open-loop gain to hover around unity instead of continuing to decrease (this is bad). The stability and step response of the op-amp will be degraded because of the zero. To illustrate this, consider the simulation results seen in Fig. 24.12. Notice that, in a transient simulation, we have to wait some time to let the bias circuit start up. Here, in this simulation, we've waited 500 ns before applying the input signal. Next, notice that our step input signal has a small amplitude change (5 mV). If we apply a larger step, the small-signal analysis used to derive pole-splitting is no longer valid (and we'll see the slew-rate limitations). Clearly the output of the op-amp doesn't behave well with the zero present. To eliminate the zero, we add the zero-nulling resistor, as seen in Fig. 21.33b. Figure 24.11 is regenerated in Fig. 24.13 with the addition of a zero-nulling resistor ($1/g_{mn} = 6.5k$). Figure 24.14 shows the well-behaved step response.

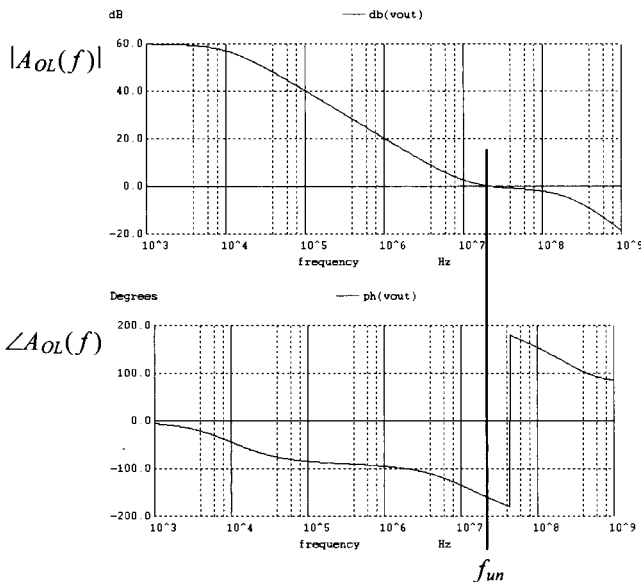


Figure 24.11 Increasing the compensation capacitor's value to 2.4 pF.

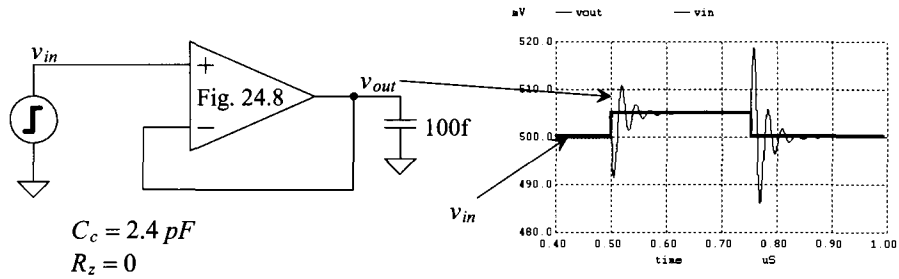


Figure 24.12 The poor step response of the op-amp with the zero present.

Gain and Phase Margins

In the previous discussions we assumed a load of 100 fF. If the load capacitance varies, the stability can (will) be affected. Further, with temperature, process, and power supply variations, the stability of the op-amp can change. In order to specify “how stable” the op-amp is at a given set of operating conditions, the parameters gain margin (GM) and phase margin (PM) are used. To determine an op-amp’s PM, we look at the phase shift when the open-loop gain is unity. The amount of phase shift away from 180° is the PM of the op-amp. As seen in Fig. 24.13, the phase shift when the gain is unity is -90° . Taking the difference between this value and 180° gives a PM of 90° . To calculate the GM, we look at the difference between the open-loop gain and unity when the phase of the op-amp is $\pm 180^\circ$. For the op-amp response in Fig. 24.13, the GM is approximately 25 dB. Note that as seen in Fig. 24.14, it is nice to have a PM of 90° because the step response of the op-amp has a first-order response (like an RC circuit).

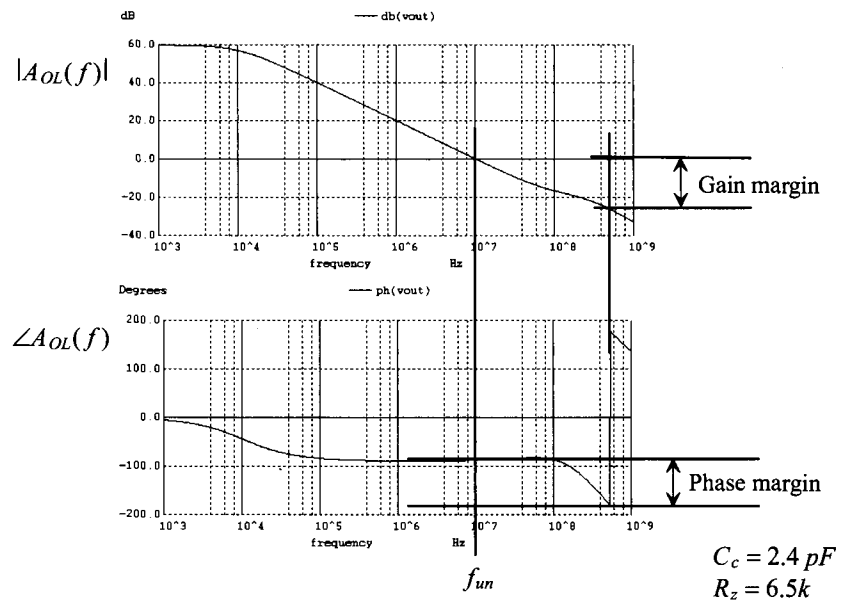


Figure 24.13 Adding a zero nulling resistor to the op-amp in Fig. 24.8.

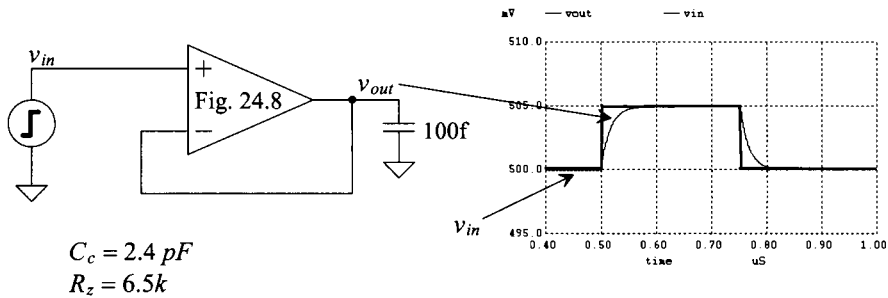


Figure 24.14 Good step response of the op-amp with the zero absent.

Removing the Zero

As the previous discussion illustrated, the RHP zero can raise issues concerning stability and settling time (this is important). As seen in Eq. (21.73) and the associated discussion, R_z can be added to eliminate ($R_z = 1/g_{m1}$) or move the zero into the LHP ($R_z > 1/g_{m1}$). When the zero is moved to the LHP, the phase response of the zero adds to the overall phase response, increasing the PM (this is called *lead compensation*). The practical problem with using R_z is that setting its value to a precise number (say $1/g_{m1}$) is challenging with shifts in process, temperature, or voltage. One solution to this problem is to replace the resistor with a MOSFET operating in the triode region, Fig. 24.15. M_z behaves like a resistor with a value of $1/g_{m1}$. Ideally, the source-gate voltage of M_7 is the same as the V_{SG} of $MP1$. It then follows that the source-gate potential of $MP2$ equals the source-gate potential of M_z . The channel resistance of M_z is then, from Eq. (9.16), set to $1/g_{m1}$. The practical issues with this method are the wasted power dissipated by the additional circuitry and the fact that if the output swing becomes large (especially at higher frequencies where C_c has a small impedance), M_z can move out of the triode region, which can affect the large-signal behavior of the op-amp.

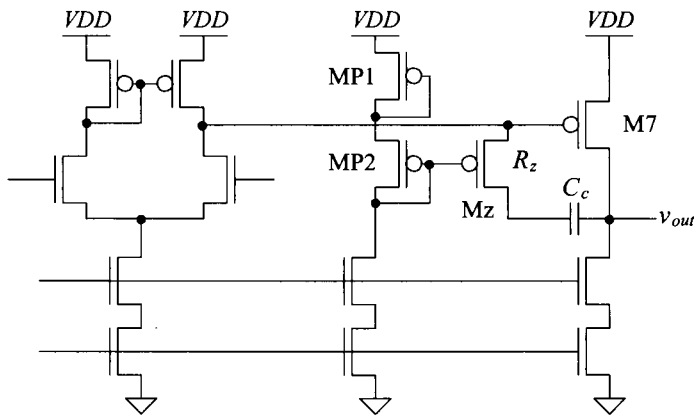


Figure 24.15 Making the zero-nulling resistor process independent.

The other method to remove the RHP zero seen in Fig. 21.33a is to add an amplifier with a gain of +1 in series with the compensation capacitor. Figure 24.16 shows the idea. A source follower allows the output signal to feed back through the compensation capacitor (so that the pole-splitting effect is still present). However, the root cause of the RHP zero, namely, C_c shunting the input of the second stage (the output of the diff-amp) to the output of the second stage (the output of the op-amp) at higher frequencies is removed. The concerns with this topology are, again, power dissipation and, perhaps more importantly, large signals. If the overall output voltage swings too low, it causes the source-follower to shut off.

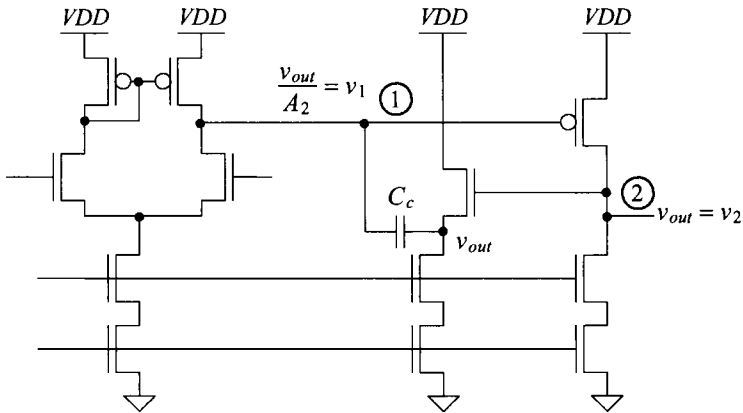


Figure 24.16 Using an amplifier to eliminate forward signal feedthrough via the compensation capacitor.

Compensation for High-Speed Operation

Notice, in Fig. 24.16 (or in any other of the topologies we've discussed up to this point), that the current fed back through C_c is

$$i_{Cc} = \frac{v_{out} - \frac{v_{out}}{A_2}}{1/j\omega C_c} \quad (24.11)$$

If the second-stage gain, A_2 ($g_{m2}R_2 = g_{m7}r_{o7}$) is reasonably large, then this equation can be approximated using

$$i_{Cc} \approx \frac{v_{out}}{1/j\omega C_c} \quad (24.12)$$

If we can feed back this current *indirectly* to the output of the diff-amp, we can still compensate the op-amp (and have pole-splitting). Further, if we do it correctly, we avoid connecting the compensation capacitor directly to the output of the diff-amp and thus avoid the RHP zero. Towards this goal, consider the modified op-amp schematic seen in Fig. 24.17. The added MOSFETs form a common-gate amplifier. (Note that here we are assuming $v_p \ll v_{out}$, so the source-follower action of MCG is negligible. MCG is connected to v_p to set its gate at the DC voltage of the input.) The current i_{Cc} is fed back through the common-gate MOSFET, MCG, to node 1 (the output of the diff-amp).

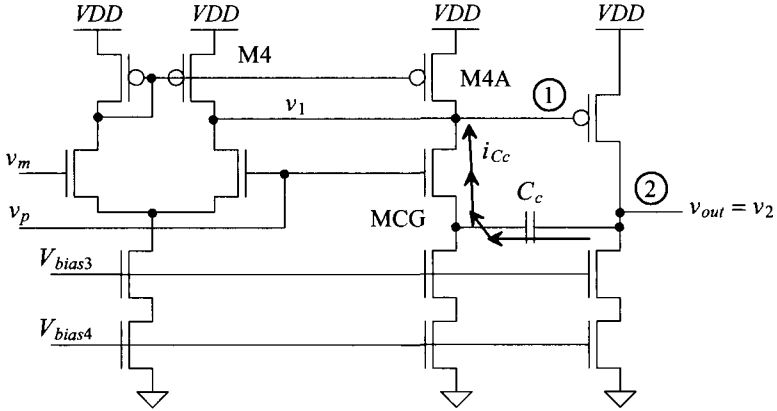


Figure 24.17 Feeding back a current indirectly to avoid the RHP zero.

To determine the frequency response of this amplifier, consider the model seen in Fig. 24.18 (modified from Fig. 21.25). Summing the currents at node 1 gives

$$-g_{m1}v_s + \frac{v_1}{R_1 \parallel \frac{1}{j\omega C_1}} - \frac{\overbrace{v_{out}}^{i_{Cc}}}{1/j\omega C_c + 1/g_{mcg}} = 0 \quad (24.13)$$

where the resistance looking into the source of MCG is $1/g_{mcg}$. Solving for v_1 gives

$$v_1 = \frac{g_{m1}R_1}{1 + j\omega R_1 C_1} \cdot \left(\frac{j\omega \frac{C_c}{g_{m1}}}{1 + j\omega \frac{C_c}{g_{mcg}}} \cdot v_{out} + v_s \right) \quad (24.14)$$

For the output node (node 2), we can write (noting that now C_2 includes both C_c and C_L)

$$v_{out} = -g_{m2}v_1 \cdot \left(\frac{R_2}{1 + j\omega R_2 C_2} \right) \quad (24.15)$$

Plugging Eq. (24.14) into this equation gives

$$v_{out} = -g_{m1}R_1g_{m2}R_2 \cdot \frac{\frac{j\omega \frac{C_c}{g_{m1}}}{1 + j\omega \frac{C_c}{g_{mcg}}} \cdot v_{out} + v_s}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)} \quad (24.16)$$

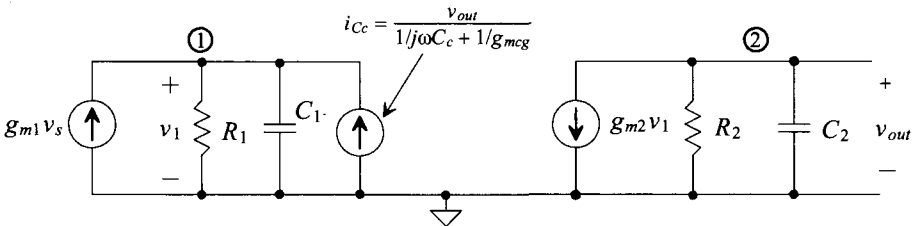


Figure 24.18 Model used to estimate bandwidth when indirect feedback current is used.

Letting

$$K = \frac{-g_{m1}R_1g_{m2}R_2}{(1+j\omega R_1C_1)(1+j\omega R_2C_2)} \quad (24.17)$$

we can rewrite Eq. (24.16) as

$$v_{out} = \frac{j\omega \frac{C_c}{g_{m1}}}{1+j\omega \frac{C_c}{g_{mcg}}} \cdot K \cdot v_{out} + K \cdot v_s \quad (24.18)$$

and, assuming $|K \cdot \frac{C_c}{g_{m1}}| \gg \frac{C_c}{g_{mcg}}$, thus

$$\frac{v_{out}}{v_s} \approx \frac{K \cdot \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{1 - j\omega \frac{C_c K}{g_{m1}}} \quad (24.19)$$

Resubstituting in K gives

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{(1+j\omega R_1C_1)(1+j\omega R_2C_2) + j\omega \frac{C_c}{g_{m1}} \cdot g_{m1}R_1g_{m2}R_2} \quad (24.20)$$

or, with $s = j\omega$, we can write

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + s \frac{C_c}{g_{mcg}}\right)}{s^2 \cdot (R_1C_1R_2C_2) + s \cdot (R_1C_1 + R_2C_2 + R_1g_{m2}R_2C_c) + 1} \quad (24.21)$$

Notice there is a LHP zero at

$$f_z = \frac{g_{mcg}}{2\pi C_c} \quad (24.22)$$

Since this zero is in the LHP, it will add to the phase response and enhance the speed of the op-amp. Intuitively, we can think that at high speeds the phase shift through C_c will cause the output signal to feed back and add to the signal at node 1. This *positive feedback* enhances the speed of the op-amp. To determine the location of the second pole, let's assume $R_1g_{m2}R_2C_c \gg R_1C_1$ or R_2C_2 or $R_1C_1R_2C_2$ so

$$s_{1,2} \approx \frac{-R_1g_{m2}R_2C_c \pm R_1g_{m2}R_2C_c}{2(R_1C_1R_2C_2)} \quad (24.23)$$

Our approximation won't tell us the location of the lower frequency pole (it's given by Eq. (21.65)). The location of the second pole is

$$f_2 \approx \frac{g_{m2}C_c}{2\pi \cdot C_1C_2} \approx \frac{g_{m2}C_c}{2\pi \cdot C_1(C_L + C_c)} \quad (24.24)$$

This result should be compared to Eq. (21.66). The location of the second pole is at a considerably higher frequency using this technique. *The result is that we can set the unity gain frequency to a higher value and still have a stable op-amp.* Further, the load capacitance (which is included in C_2) can be considerably larger for a given PM or GM. Again, Eq. (21.72) can be used to set the unity-gain frequency, f_{un} , assuming $f_2 \approx f_z$

$$f_{un} = \frac{g_{m1}}{2\pi C_c} (\approx f_z \text{ if } g_{m1} \approx g_{mcg}) \quad (24.25)$$

Using the *indirect compensation* method seen in Fig. 24.17 in the op-amp of Fig. 24.8 with $C_c = 240 \text{ fF}$ ($f_{un} = 100 \text{ MHz}$) gives the results seen in Fig. 24.19. The small-signal step response is seen in Fig. 24.20. This response should be compared to Figs. (24.12) and (24.14).

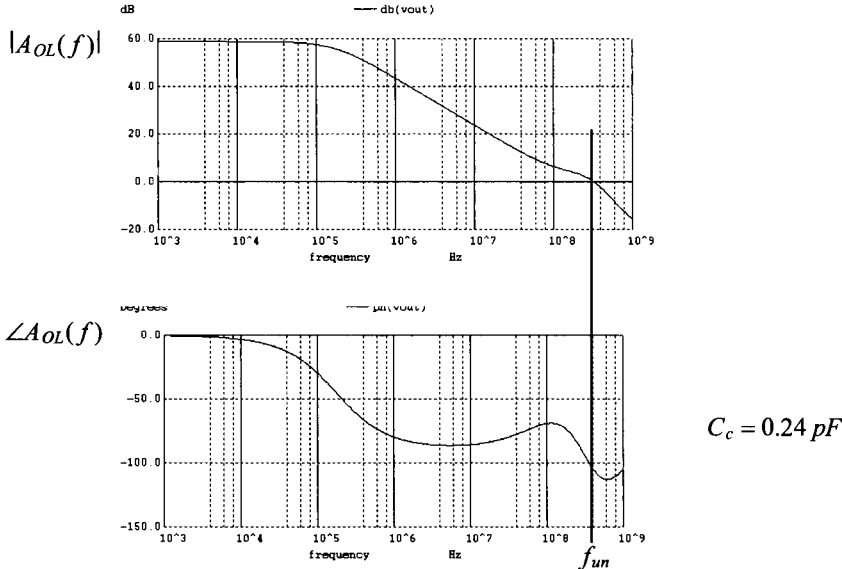


Figure 24.19 Simulating the op-amp of Fig. 24.8 using the indirect compensation scheme seen in Fig. 24.17 with a compensation capacitor of 240 fF.

The indirect feedback of the current through the compensation capacitor results in faster op-amp circuits and less layout area (the compensation capacitor generally dominates the layout area of an op-amp). However, the added circuit in Fig. 24.17 dissipates more power. To eliminate the additional power dissipation, consider the op-amp topology seen in Fig. 24.21. Here we've used the fact that a 100/2 PMOS device can be laid out as two 100/1 PMOS in series (see also Fig. 20.35). The current through the compensation capacitor is fed back through M4B to the output of the diff-amp. Note how we've also split M7 into two devices. It's important, for small offsets, to try to match both the drain-source and the gate-source voltages of the MOSFETs used in the op-amp. The step-response of the op-amp is seen in Fig. 24.22.

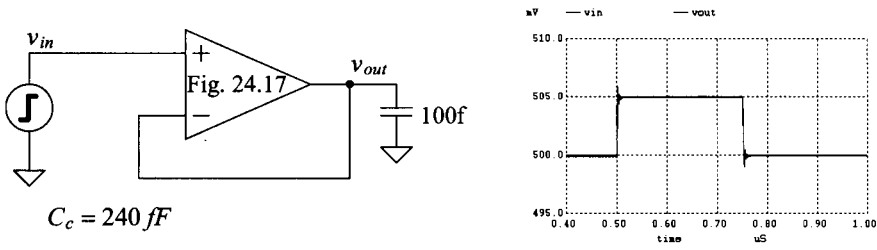


Figure 24.20 The step response of the op-amp in Fig. 24.17 with frequency response seen in Fig. 24.19.

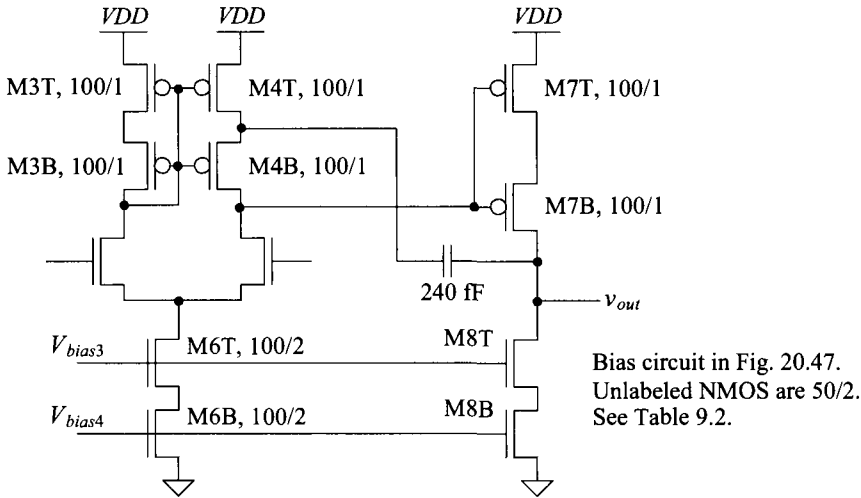


Figure 24.21 Implementing indirect feedback compensation without additional power dissipation in a two-stage op-amp.

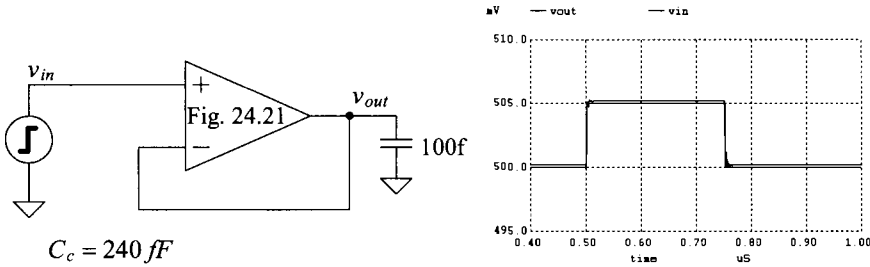


Figure 24.22 The step response of the op-amp in Fig. 24.21 driving 100 fF.

Slew-Rate Limitations

When we talked about step response, we limited the amplitude of the signals to small steps (5 mV) to avoid slew-rate limitations. Let's take the op-amp from Fig. 24.8 and put it in the configuration seen in Fig. 24.14, driving a 1 pF load capacitance. Further, let's increase the input pulse amplitude to (close to) the maximum allowable range, that is, from 500 mV to 900 mV (limited by the op-amp's input common-mode range). The simulated results are seen in Fig. 24.23. Referring to Fig. 24.8, notice that when the op-amp's noninverting input terminal (v_p) is driven high, M2 turns on and pulls the gate of M7 down. Since there isn't a current source in series with M7, it can quickly charge the 1 pF load capacitance. The only large-signal limitation is the bias current of the diff-amp, I_{SS} ($= 20 \mu\text{A}$ here), charging C_c ($= 2.4 \text{ pF}$ here). This limits the output rate of change since any variation in v_{out} causes a displacement current through C_c , which must be sourced/sunk by the diff-amp. This slew-rate limitation is calculated as $20 \mu\text{A}/2.4 \text{ pF}$ or 8.3 mV/ns. The change in the output voltage is 400 mV, so we would require roughly 50 ns to change the voltage across C_c .

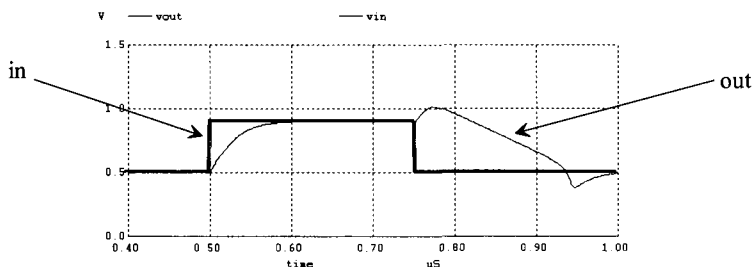


Figure 24.23 The large-signal (500 mV to 900 mV) performance of the op-amp in Fig. 24.8 with $C_c = 2.4$ pF and $R_z = 6.5$ k driving a 1 pF load. The low-to-high settling time is roughly 100 ns, while the high-to-low settling time, which is slew-rate limited, is roughly 500 ns.

The more interesting behavior occurs when the input signal drops from 900 mV to 500 mV. This causes M2 to shut off, allowing all of the current from M4 to charge the gate of M7. The result is a positive pulse on the gate of M7 that feeds directly to the output through the compensation capacitor (this is bad, and it is another example of why it is desirable to use the indirect compensation method). This pulse accounts for the (unwanted) positive movement in the output voltage seen in Fig. 24.23 just after the input pulse transitions negative. Now the load capacitor and the compensation capacitor must both be discharged through the constant current source M8. The output slew-rate is estimated by how fast this constant current (here 10 μ A) can discharge 3.4 pF (the sum of the compensation capacitor and the load capacitor). This rate is calculated as (roughly) 3 mV/ns. For the output to transition 400 mV requires 133 ns (this time will be longer because of the positive movement in the output voltage just after the input switches). Note that we might think that simply by sizing up the current conducting in M8 we can enhance the op-amp's slew-rate. This is true to the point where the diff-amp's current charging the compensation capacitor becomes the limiting factor.

Finally, note how the output voltage shoots past the final (desired) voltage level of 500 mV. This is because the gate of M7 is pulled to V_{DD} to shut it off when slewing is taking place (both M4 and M7 are shut off because of this). When the inputs of the op-amp move to the same value (500 mV here), the gate of M7 must be pulled downwards to the quiescent value. The time it takes for this to happen results in the undershoot in the op-amp's output voltage seen in Fig. 24.23.

The same test setup used to generate the data in Fig. 24.23 is also used to generate the data in Fig. 24.24 except that here the op-amp in Fig. 24.21 is used. Note the compressed time scale. The settling time for the low-to-high transition is now 10 ns, while the high-to-low transition time is 60 ns. We would expect much faster settling because the compensation capacitor is now ten times smaller than the value used in the op-amp of Fig. 24.8. Again, the settling time is limited by how fast M8 can discharge the load capacitance and the compensation capacitance. This can be estimated as $10 \mu\text{A}/1.24 \text{ pF}$ or 8 mV/ns. It takes 50 ns for the output to transition 400 mV. The size of M8 can be increased to meet a specific load capacitance and settling time requirement.

Because of the improved speed performance, smaller layout area, and (as we'll see) better power supply noise rejection, we'll use indirect compensation in place of Miller compensation in the remaining two-stage op-amps that we discuss in this book.

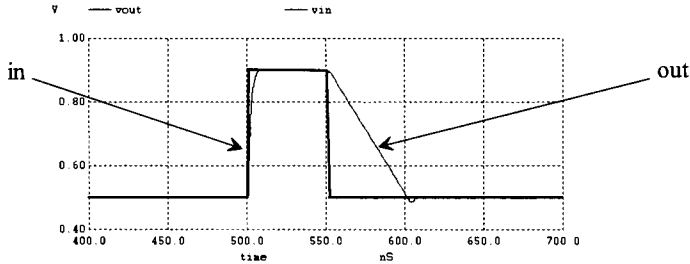


Figure 24.24 The large-signal (500 mV to 900 mV) performance of the op-amp in Fig. 24.21 with $C_c = 0.24$ pF driving a 1 pF load. The low-to-high settling time is roughly 10 ns, while the high-to-low settling time, which is slew-rate limited, is roughly 60 ns. Note the different time scale when compared to Fig. 24.23.

Common-Mode Rejection Ratio (CMRR)

The *CMRR* of an op-amp is calculated in the same way as the diff-amp in Sec. 22.1.3. The common-mode gain of the diff-amp is A_c . The common-mode gain of the op-amp is $A_c A_2$. The differential gain of the op-amp is $A_{OL}(f) = A_d \cdot A_2$ (where $A_d = A_1$). The *CMRR* of an op-amp in dB is given by

$$CMRR = 20 \cdot \log \left| \frac{A_{OL}(f)}{A_c \cdot A_2} \right| = 20 \cdot \log \left| \frac{A_d}{A_c} \right| \quad (24.26)$$

which shows that the op-amp *CMRR* is determined by the differential stage. Simulating the *CMRR* of an op-amp can be accomplished with the circuits seen in Fig. 24.25. For the op-amps discussed so far in this chapter, the *CMRR* is approximately 50 dB (see Ex. 22.7, where the common-mode voltage is 500 mV). We can think of A_c as the open-loop gain of the op-amp for common-mode signals. If the applied differential voltage is zero, and we change the common-mode voltage by ΔV_c , then the output voltage will change by ΔV_o .

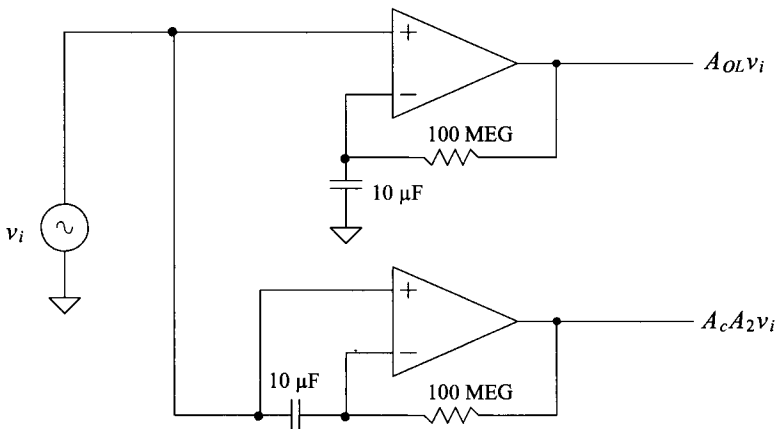


Figure 24.25 Circuit configuration used to simulate *CMRR*.

$= A_{cm} \cdot \Delta V_c$ (see Fig. 22.17). To compensate for the change in the output voltage, a nonzero input differential voltage develops on the input of the op-amp (an offset voltage that is a function of the common-mode voltage). This offset voltage can be estimated by

$$\Delta V_{OS} = \frac{\Delta V_o}{A_{OL}} = \frac{\Delta V_c \cdot A_{cm}}{A_{OL}} = \frac{\Delta V_c}{CMRR} \quad (24.27)$$

Knowing that 50 dB = 316, a change in the common-mode level on the inputs of the op-amp by 500 mV results in an input-referred offset voltage change of 500 mV/316 or 1.6 mV (if the gain of the diff-amp is 10, then its output voltage will change by 16 mV). This may not seem like such a big deal. However, notice that at higher frequencies, in Fig. 22.16, the $CMRR$ falls off, indicating that the common-mode gain is increasing. This leads to distortion and forces the use of op-amp topologies in which common-mode voltage doesn't vary (the inverting op-amp topology, see Fig. 24.26).

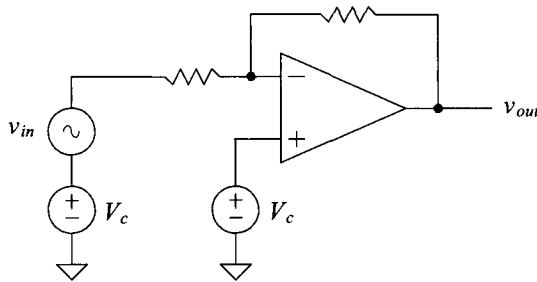


Figure 24.26 An inverting op-amp topology. The common mode voltage is held constant.

Power Supply Rejection Ratio (PSRR)

The power supply rejection ratio (PSRR) is a term used to describe how well an amplifier rejects noise or changes on the V_{DD} and or ground power buses. This parameter can be extremely important in precision analog design. Consider the test setup shown in Fig. 24.27. The positive PSRR is defined by

$$PSRR^+ = \frac{A_{OL}(f)}{v_{out}/v^+} \quad (24.28)$$

while the negative PSRR is defined by

$$PSRR^- = \frac{A_{OL}(f)}{v_{out}/v^-} \quad (24.29)$$

Ideally, v_{out} doesn't vary with changes in V_{DD} and ground (and so the $PSRR$ is infinite).

To understand how variations in V_{DD} or ground can feed to the output of the amplifier, consider the op-amp in Fig. 24.8. Variations in V_{DD} feed through to the gate of M3. Because of the circuit's symmetry, this causes the drain of M4, and thus the gate of M7, to move around. However, the source of M7 is moving the same amount, causing the v_{sg} of M7 to remain constant. Again, because of the symmetry, the output voltage moves